Lista 0 - Processo adaptativo de sinais

Elaborar um script em Matlab ou Python que reproduza as Figuras 2.8 e 2.9 relativas ao exemplo 2.5 do livro do Diniz 5 ed.

Example 2.5 The matrix **R** and the vector **p** are known for a given experimental environment:

$$\mathbf{R} = \begin{bmatrix} 1 & 0.4045 \\ 0.4045 & 1 \end{bmatrix}$$

$$\mathbf{p} = [0 \ \ 0.2939]^T$$

$$\mathbb{E}[d^2(k)] = 0.5$$

(a) Deduce the equation for the MSE.
(b) Choose a small value for μ, and starting the parameters at $[-1 \ \ -2]^T$ plot the convergence path of the steepest descent algorithm in the MSE surface.
(c) Repeat the previous item for the Newton algorithm starting at $[0 \ \ -2]^T$.